

TAN ZHENG XUN

+65 8111 0717 | tzx8787@gmail.com | [linkedin.com/in/tan-z-x](https://www.linkedin.com/in/tan-z-x) | <https://github.com/tzx21100>

EDUCATION

Digipen Institute Of Technology Singapore

Sep 2022 - May 2026

- **Bachelor of Science in Computer Science in Interactive Media and Game Development**
- GPA: 4.3/5.0
- **Hackathon: CoDE 2023 Runner Up**

EXPERIENCE

Venti Technologies

Apr 2025 - Apr 2026

Simulation Engineer Intern

- Designed and maintained a large-scale autonomous-vehicle simulation framework supporting complex multi-agent scenarios with strict structural and execution constraints.
- Built an AI-powered scenario generation tool by fine-tuning open-source LLMs with PyTorch, deploying local inference via llama.cpp, and integrating tool-based workflows to automate simulation scenario creation.
- Worked extensively in Linux environments, applying defensive programming practices in Python and C++.

Redalpha Cybersecurity

Jun 2026 - Sep 2026

CCTP Cybersecurity Trainee

- Completed intensive hands-on training in penetration testing, malware analysis, reverse engineering, digital forensics and Windows/network security.
- Investigated software and systems using Ghidra, Burp Suite, Wireshark, Volatility, Sysmon and debugging/analysis tooling.

PROJECTS

INSIGHT ENGINE | Custom 2D Game Engine, Team of 6

Apr 2024 - Sep 2024

Technical Lead | Programmer

- Led a 6-person team developing a custom C++ 2D engine with ECS, multithreading, memory management and Mono C# scripting.
- Engineered an ECS supporting 1M+ entities and a multithreaded asset pipeline reducing load times by 76%.
- Implemented custom memory management and prefab, input, audio and messaging systems

IRON ENGINE | Custom 3D Game Engine, Team of 12

Jan 2025 - April 2025

Systems Programmer

- Developed core systems for a custom C++ 3D engine using EnTT and RTTR.
- Designed a behaviour-tree system and runtime-reflection infrastructure supporting designer-facing gameplay workflows.

Combatplus | Turn-Based Combat Overhaul Mod (Stardew Valley)

May 2026 - Present

Personal Project

- Developed and published a multiplayer-compatible combat overhaul that transformed Stardew Valley's real-time combat into a turn-based system.

SKILLS & ACHIEVEMENTS

Language: C, C++, C# Mono/.NET, Python, Kotlin

Tools & Frameworks: Unity, Mono C#, Git, Linux(Ubuntu), Llama.cpp, Pytorch

System & Design patterns: Memory management, multi-threading, ECS, performance profiling

Security: Ghidra, Burp Suite, Wireshark, Volatility 3, Sysmon, Sysinternals, Nmap, x64dbg

AI/ML: PyTorch, llama.cpp, local LLM inference & fine-tuning